

# Machine Learning-Enhanced Acoustic Reflectometry for Early Detection of Middle Ear Infections

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**Middle ear infections are a prevalent health issue affecting individuals across all age groups. Timely and accurate detection is crucial for effective treatment and preventing complications. This study introduces an innovative smartphone-based machine learning system for detecting middle ear infections. The system combines an acoustic analysis module, using acoustic reflectometry theory, with a custom smartphone application. By connecting an earphone device to the smartphone, chirping noises are emitted and received in the ear, and the embedded microphone captures and analyzes the reflected sound waves using a machine learning algorithm. The system accurately categorizes the results into "no water," "monitor," and "water," identifying middle ear fluid, an important indicator of infection. Early detection of fluid buildup can lead to timely intervention, potentially preventing infections. Extensive testing was conducted on various subjects to validate the system's accuracy against a commercial device. The research represents a significant advancement, providing a non-invasive, accessible, and cost-effective solution for detecting middle ear infections in both home and clinical settings, promoting self-monitoring of ear health, and reducing associated complications.**

**Index Terms**—Middle ear infection, Middle ear effusion, Acoustic reflectometry, Machine learning

## I. INTRODUCTION

Middle ear diseases, such as acute otitis media (AOM) and otitis media with effusion (OME), are prevalent pediatric health concerns. AOM, commonly known as an "ear infection," exerts a significant healthcare burden, contributing to substantial costs and a high number of office visits in the United States [1]. These infections often arise as complications of colds, flu, allergies, or conditions leading to nasal congestion, introducing bacteria or viruses into the middle ear and causing fluid buildup in the air-filled space behind the eardrum (Tympanic cavity).

OME, affecting up to 80% of children, presents with middle ear fluid without acute infection signs [2]. Despite minimal symptoms, OME can lead to speech delays, disrupted sleep, poor school performance, balance issues, and an increased risk of developing AOM [2].

Diagnosing AOM or OME typically involves detecting middle ear fluid likelihood through methods like pneumatic otoscopy or tympanometry. However, these approaches have limitations, with pneumatic otoscopy being underused and not suitable for home screening, while tympanometry requires specialized audiologists and expensive equipment [1,3].

To address these challenges, our system integrates Acoustic Reflectometry theory to assess eardrum mobility and detect middle ear fluid. Utilizing modified earphones, we send a gentle acoustic chirp into the ear canal, detecting and analyzing the reflected sound waves using our connected system based on Acoustic reflectometry theory. Then a machine learning model is employed to classify these reflections and predict the middle ear fluid status accurately.

Our smartphone-based solution aims to revolutionize ear infection detection, providing cost-effective accessibility and an efficient means to identify middle ear fluid. Through early detection, users can proactively identify potential issues and schedule a doctor's visit promptly, reducing the likelihood of complications. By empowering individuals with timely information from the device's results, we aim to enhance the management of ear health and minimize the risks associated with these common ear diseases.

To validate the system's accuracy and reliability, extensive clinical studies and further research are essential. The integration of Acoustic Reflectometry instills confidence in the robustness of our machine learning model in effectively classifying eardrum reflections related to middle ear fluid.

Our smartphone-based system represents a promising advancement in ear infection detection and management. With the potential to enhance the health of countless children, this technology facilitates early intervention, lessening the overall impact of these prevalent ear conditions.

## II. PRINCIPLE OF ACOUSTIC REFLECTOMETRY

Acoustic Reflectometry (AR) is a non-invasive technique used to assess the likelihood of Otitis media with effusion (OME) by analyzing how the tympanic membrane responds to sound stimuli with specific frequencies. AR offers reliability even with crying or struggling children as it does not require an airtight seal in the ear canal [4].

The stapedius reflex, also referred to as the acoustic reflex, is an automatic contraction of the intra-aural muscle in the middle ear, triggered by exposure to loud sounds. This reflex is bilateral, occurring simultaneously in both ears, even if the loud sound only reaches one ear. Through acoustic reflectometry, we can record and evaluate the acoustic reflexes, including the acoustic reflex threshold, which denotes the minimum sound volume required to elicit the stapedius muscle's contraction.

Typically, the acoustic reflex threshold falls within the range of 65 to 90 dB [5].

In acoustic reflectometry, sound stimuli with specific frequencies, such as 500, 1000, 2000, and 4000 Hertz (Hz), along with broadband noise, are employed. The concept of reflectivity is utilized to assess the extent to which the incident sound waves from the instrument are counteracted by the reflecting sound waves. The degree of cancellation is influenced by factors like the phase and frequency of the emitted and reflected sound waves. A more advanced technique, known as spectral gradient acoustic reflectometry (SG-AR), graphically presents data derived from both incident and reflected sound. This enables the determination of the spectral gradient angle, which offers valuable insights into the tympanic membrane's response to sound stimuli within the frequency range of 1.8-4.4 kHz[6]. A narrow spectral gradient angle indicates a higher level of sound energy reflection and a narrower frequency spectrum, potentially indicating constrained membrane movement due to the presence of middle ear fluid. The classification of the spectral gradient curve into various pattern categories can then provide information about the likelihood of Middle Ear fluid [1].

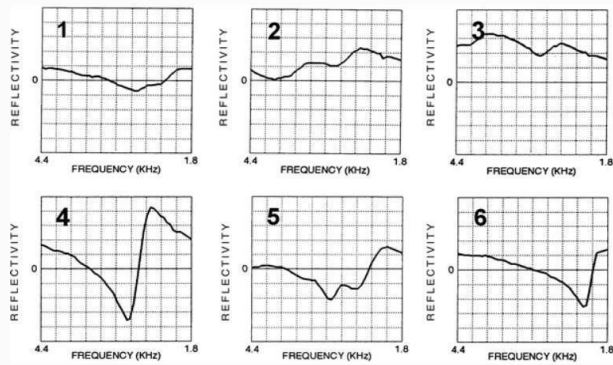


Fig. 1: The image illustrates six different acoustic reflects curve types, each representing a varying probability of the presence of middle-ear fluid in the examined ear. The middle ear without middle ear fluid generally falls into patterns 1 to 3. Pattern 4 shows the presence of middle ear fluid as the curve has a narrow spectral gradient angle. Patterns 5 and 6, indicate a rightward shift of the spectral gradient angle or a shift towards lower frequency, respectively, are associated with marginally normal cases, negative pressure, or a minor amount of middle-ear fluid [1].

### III. METHODOLOGY

#### A. Acoustic reflectometry system

The objective of our research is to develop and validate an acoustic reflectometry system for accurate detection of middle ear fluid levels, focusing on tympanic membrane mobility. The system consists of a modified earphone housing a microphone and a speaker, which emits soft acoustic chirps into the ear. The system's microphone captures the reflected sound waves, which are subsequently analyzed using machine learning algorithms to interpret the acoustic dip resulting from the sound waves' interaction with the eardrum.

A normal eardrum resonates effectively, resulting in a broad-spectrum, soft echo and a corresponding broad and shallow acoustic dip. Conversely, in ears with middle ear effusion (MEE), the fluid impedes the eardrum's mobility, leading to

reflected sound energy causing destructive interference, resulting in a narrower and deeper acoustic dip.

To achieve precise data collection and analysis of the acoustic dip, we designed and constructed multiple systems, including the modified earphone for sound emission and collection, an artificial tympanic cavity as a substitute for the human ear, and advanced machine learning algorithms for acoustic dip analysis.

#### B. Development and Construction of the Earphone.

The acoustic reflectometry system incorporates a customized earphone for its functioning. This earphone emits frequency-modulated continuous wave chirps at 150 MS with frequencies ranging from 1.8 to 4.4 kHz into the patient's ear canal. Concurrently, the active microphone, integrated within the modified earphone, captures both the incident and reflected sound waves emanating from the tympanic membrane during the emission of the acoustic chirp.

The earphone's design was designed using Inventor® CAD software and printed 3D printing technology. The device comprises two essential components: the body, housing the closely positioned speaker and microphone, and the replaceable tips, facilitating easy hygienic cleaning and sizing adjustments. This modified earphone easily connects to computers or smartphones through a standard 3.5mm connection.

The development of this specially tailored modified earphone significantly enhances the precision and efficiency of the acoustic reflectometry system in accurately detecting middle ear fluid levels.



Fig. 2: The earphone on the left-hand side illustrates the inner circuit system of the earphone which is composed of microphone and speaker. On the right-hand side is shown the completed earphone model with removable tips.

#### C. Development and Construction of the artificial tympanic cavity.

For the purpose of acoustic reflects data collection for machine learning analysis, sounds reflect from middle ear samples are required. However, conducting direct data collection from actual human subjects poses ethical challenges, and obtaining a large sample size with varying levels of fluid in the middle ear can be difficult. To overcome these obstacles, an artificial tympanic cavity is developed as a controlled

environment for data collection, eliminating ethical concerns and allowing for precise and controlled trials.

The artificial tympanic cavity is meticulously designed to closely resemble the human middle ear structure. It consists of a 3D printed tympanic cavity with a volume of 1ml, accurately replicating the anatomical features of the human ear. A silicone tube, similar in length to the human ear canal, is attached to the cavity, serving as the ear canal through which the acoustic chirp enters [7].

To simulate the tympanic membrane, a Parafilm® layer is placed between one end of the silicone tube and the open side of the artificial tympanic cavity. This Parafilm® membrane imitates the acoustic properties of the human tympanic membrane. Additionally, the artificial tympanic cavity features an open top that can be filled with water, mimicking different levels of fluid commonly found in real human ears with middle ear effusion.

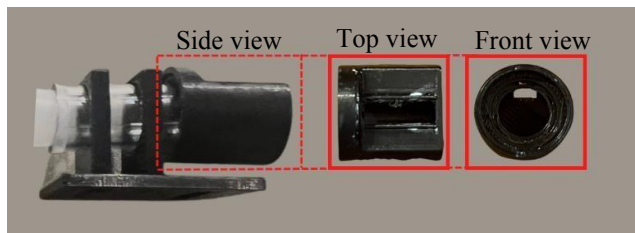


Fig. 3: Different angles illustrate the artificial tympanic cavity.

#### D. Validation of the Artificial Tympanic Cavity: Comparing Performance with the Earcheck™ Commercial Devices.

Validating the performance of the artificial tympanic cavity is crucial to ensure accurate data collection and minimize potential errors. By testing its performance with the well-established Earcheck™ commercial device. Earcheck™ device is the commercial acoustic reflectometry device. According to the study evaluated the effectiveness of consumer-type acoustic reflectometry, specifically the Earcheck™ device, in detecting otitis media with effusion (OME) compared to tympanometry. The results showed that Acoustic reflectometry demonstrated a sensitivity of 83.33% and specificity of 68.18%, with a positive predictive value (PPV) of 75.12% and negative predictive value (NPV) of 87.66%. Overall, the study concludes that consumer-type acoustic reflectometry device, is a highly sensitive and objective technique for detecting OME. It provides an easy-to-use approach that can be employed by family practitioners or parents, enabling early-stage diagnosis of OME. Despite having slightly lower specificity and predictive values compared to tympanometry [4].

The artificial tympanic cavity is designed to closely mimic the structure and conditions of the human middle ear, ensuring accurate detection of middle ear fluid levels. To achieve this, a series of controlled experiments are conducted, using the artificial tympanic cavity and the Earcheck™ device. The device measures the amount of middle ear fluid in a human ear using the theory of acoustic reflectometry and categorizes the results into five levels: “Fluid unlikely”, “monitor”, “consult doctor level 3”, “consult doctor level 4”, and “consult doctor level 5”,

with the severity level increasing with the amount of water detected.

The results from our experiments demonstrate that the artificial tympanic cavity closely emulates the functioning of a real human ear, replicating the characteristics of acoustic reflectometry observed in typical human ears by using the a commercial device. As water is added to the artificial tympanic cavity, the indications from the device show a clear parallel, with "Fluid unlikely" when no water is present, "Monitor" as water is gradually added, and "Consult doctor" at a specific volume of water.

#### E. Data collecting.

The acoustic reflectometry data collection and preprocessing involved several steps. Initially, we connected the Earcheck™ acoustic reflectometry device to our artificial tympanic cavity. By varying the amount of water inside the cavity, we conducted scans with the commercial device to determine the water level. Subsequently, using our modified earphone, we played a 150-ms chirping noise into the same artificial tympanic cavity with the corresponding water level. The microphone recorded the incident and reflected waves as a .wav file. This process was repeated 75 times, creating five cohorts of different water levels, each containing 15 datasets.

Comparing the data obtained from our artificial middle ear system to the results recorded by the commercial acoustic reflectometry device allowed us to observe the appearance of acoustic waves associated with each water level. This enabled us to organize the data into categories similar to those obtained from the commercial equipment, ensuring accurate comparison and analysis.

#### F. Data preprocessing.

The acoustic chirp data consisted of 7200 samples that were transmitted to the middle ear system. Upon recording, a total of 48,000 samples were obtained. To extract the frequency response, we applied a fast Fourier transform (FFT) to the .wav file, focusing on the frequency range of 0-24 kHz. Subsequently, we filtered out frequencies outside the 0-5000 kHz range of the transmitted chirp. This specific range was chosen to isolate the acoustic dip area of the graph, which is essential for conducting accurate machine learning analysis.

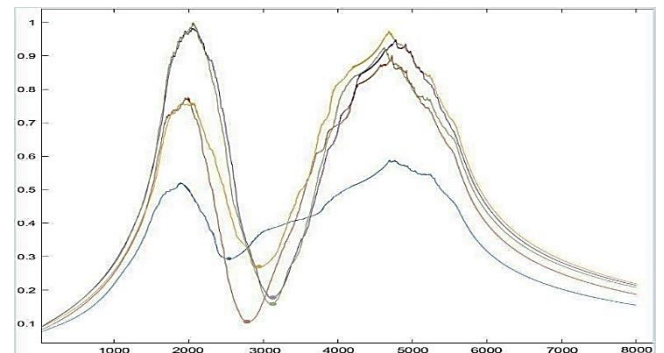


Fig.4: Different levels of acoustic wave form obtained when chirps were played into an ear without fluid (blue), with minimal fluid (yellow), with medium fluid (purple and green) and maximum fluid (orange).

To prepare the 75 acoustic response data for machine learning training, range extraction was performed on all the data before classification. For training, there are two separate spreadsheets.

All five cohorts used for AI training are included in the first spreadsheet, dataset A. The term "Consult doctor" is used to classify cohorts that range from "Consult doctor level 3" to "Consult doctor level 5". Consult doctor referring to that there is a certainty of fluid inside the middle ear. Different levels of consult doctor are associated with the expected amount of water inside, level 5 referring to the most fluid likely inside the ear. So, together, there will be 3 categories of water level: "Fluid unlikely", "Monitor", and "Consult doctor", each with 15 data points, including 5 from each "Consult doctor" level segment, making a total of 15 data points.

The machine learning model learns from the categorized data in these spreadsheets, which act as the training dataset, enabling accurate classification and analysis.

#### IV. RESULTS

After importing dataset, and into MATLAB's classification learner, we conducted k-fold cross-validation ( $k = 5$ ) to evaluate various machine learning models. Three models, namely Cubic Support Vector Machine (SVM), Tri-layered Neural Network, and Fine K-Nearest Neighbors (KNN), demonstrated the highest accuracy during testing. The accuracy rates achieved by these models were 80%, 80%, and 76.7%, respectively.

#### V. CONCLUSION & DISCUSSION

In conclusion, our research has successfully developed a middle ear screening tool by integrating acoustic reflectometry theory into a system comprising an artificial tympanic cavity, modified earphone, and chirping noise. We utilized machine learning in MATLAB, yielding promising results. However, it is essential to acknowledge the limitations of our data gathering technique, which focused on the artificial tympanic cavity and may have introduced biases. To enhance the accuracy and applicability of our technology, future studies should obtain real human data after obtaining relevant ethical clearances.

The potential impact of our middle ear screening system, based on acoustic reflectometry theory, is significant for early detection of middle ear effusion (MEE), by incorporating real human data, we can address potential biases, improve the reliability and generalizability of our findings, and better adapt the system to real-world complexities. This study represents a crucial step towards practical MEE screening using acoustic reflectometry and has the potential to revolutionize in the hearing healthcare department.

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